

GURANSH SINGH KOHLI

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EDUCATION

University of Waterloo

Sept 2025 - April 2030

Bachelor of Honours Mathematics Coop

Double major in Scientific Machine Learning, Statistics & Computing Minor

SKILLS

Languages	C, Python, Java, JavaScript, GO ,SQL,DrRacket, TypeScript, HTML,CSS
Frameworks & Libraries	React,Node.js,Express.js, FastAPI, Sci-kit Learn
Systems & Tools	AWS, MySQL, Git, Github, MongoDB, Docker, Bazel
Core CS/ML	Pandas, NumPy, PyTorch, EDA, Excel, Tableau, Power BI, Matplotlib, Seaborn

EXPERIENCE

Savi Finance

Dec 2025 – Present

AI & Software Developer

Waterloo, Canada

- Built and integrated **backend services** and **automation components** to streamline internal workflows.
- Prototyped, tested, and debugged **production-facing features** supporting **1.5k+ active users**.
- Collaborated on **scalable software solutions** to improve system reliability in an early-stage **fintech** environment.

Wanderers – Community for Explorers

Jan 2025 – Present

Co-Founder, Startup

Waterloo, Canada

- Co-founded and engineered a **community platform** enabling **interest-based user discovery**.
- Implemented core **backend and application features** using **user profiles, tagging, and matching logic**.
- Designed **scalable backend pipelines** to support **future recommendation systems** and personalization.

WUSA – Waterloo Undergraduate Student Association

July 2025 – Present

Ownership Consulting Committee Member (At-large)

University of Waterloo, Canada

- Represented **40,000+ students** across campuses through **board-level discussions**, voicing concerns and contributing to solutions.
- Collaborated with **cross-functional committees** to **evaluate and improve student services** across campuses.

PROJECTS (GITHUB.COM/GURANSHSINGHKOHLI)

ChurnLens — Customer Churn & Revenue Analytics Platform

- **Objective:** Design a **scalable data processing system** to analyze **customer churn** and quantify **revenue impact** for data-driven product decisions.
- **Features:** Built **modular, optimized SQL pipelines** using **CTEs and JOINS** to compute churn metrics, customer lifetime value, and revenue loss, supporting **maintainable and reusable workflows**.
- **Skills:** **SQL** (CTEs, JOINS, CASE), **Data Modeling, Pipeline Design, Query Optimization, Tableau, Power BI**

Accident Severity Predictor — Data Visualization & Machine Learning System

- **Objective:** Build an end-to-end ML workflow to predict **accident severity** and analyze **city-level trends**.
- **Features:** Developed data ingestion and **EDA pipelines**; applied **K-Means clustering** for pattern discovery and trained a **supervised ML model** achieving **80% accuracy**.
- **Skills:** **Python, MySQL, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn**

VectorMate — Neural-Network-Based Chess Engine (In Progress)

- **Objective:** Build a **neural-network-powered chess engine** with a target **ELO rating of 1000+**.
- **Features:** Developed a **chess bot** with **neural evaluation functions** to improve **move selection**, search quality, and overall gameplay strength.
- **Skills:** **Python, PyTorch, Neural Networks, Algorithmic Search, Node.js, Problem Solving**